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Capstone Reflection
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EEE 4775 Summer 26

What was harder than expected:

For me, the most challenging part was getting down the exact timing differences between the binary semaphores and direct task notifications. Because of how the ESP32 reads from memory, the latency was jumping all over the place. I used the IRAM_ATTR flag to force the code to run directly on RAM before the numbers leveled out and became predictable. Trying to balance the background tasks so they didn't ultimately starve the CPU and crash everything took some trial and error as well.

What I would do differently:

If I had to do this app again or make a real product to sell in the real-world, I would definitely change how the button is set. Right now I'm using software to filter out any noise when the button is pressed but this ultimately wastes CPU power. In real life I would use hardware components on a circuit board to filter out the noise. Also, like I mentioned in my demo video at the end, I would switch to just direct task notifications rather than semaphores because for what is needed here they just run a lot faster and also use less memory.

The most valuable thing I learned:

The most valuable thing I learned this semester was seeing how task scheduling works in practice rather than just knowing what it is and what it means. Seeing how assigning different priorities based on how often a task runs kept the whole system stable even when the CPU was essentially maxed out. Adding the fault injection also was cool to see because it showed me that having a good system isn't just having everything run and work; it's about error and hazard management as well. When things fail and can be potentially unsafe in real applications, you need to know that this is just as important as things working perfectly when conditions are stable.